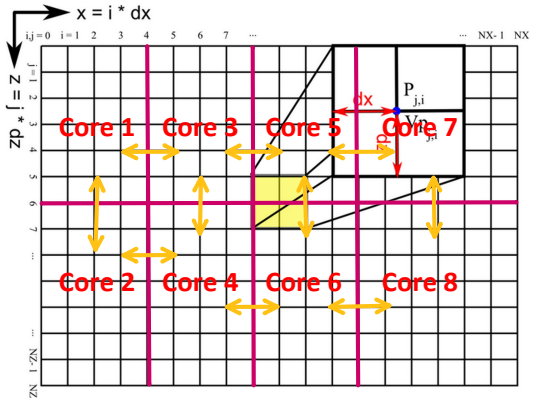
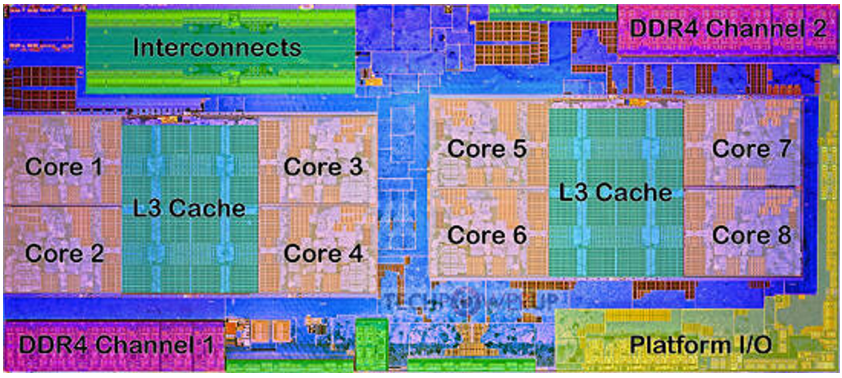
Code Performance Improvements

Sonntag, 24. Mai 2020 16:45

1. Domain Decomposition using the Message-Passing Interface (MPI)





Pressure wavefield exchange at domain boundaries using MPI

```
# FD approximation of spatial derivative by 3 point operator
for i in range(1, nx - 1):
    for j in range(1, nz - 1):

        d2px[i,j] = (p[i + 1,j] - 2 * p[i,j] + p[i - 1,j]) / dx ** 2
        d2pz[i,j] = (p[i,j + 1] - 2 * p[i,j] + p[i,j - 1]) / dz ** 2
```

2. Partioning of the tasks in each domain using Multithreading (OpenMP)